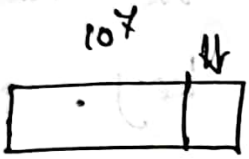
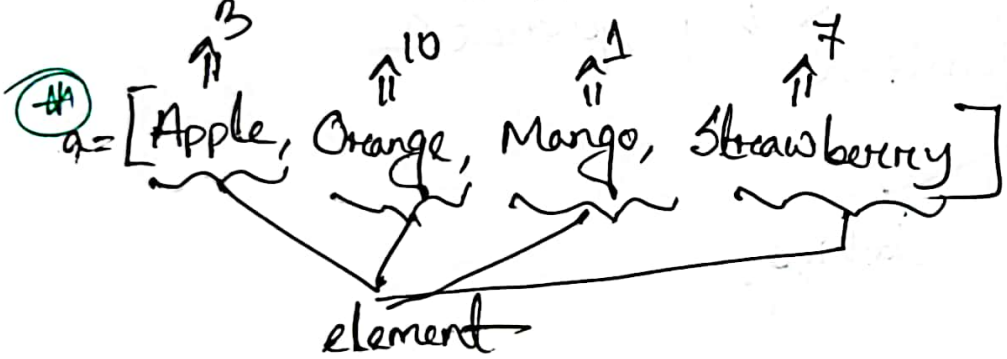
02-08-23

HASHING

Search
- Binary → Sorted
- Linear → take too much time.



$10^7 - 1$ → index.



each element → mapped to a key (specific)

dict = {3: Apple, 10: Orange, 1: Mango, 7: Strawberry}

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(#) Element of an array ⇒ map key

Hashing.

(#) Hash function → the function we need to generate a key.

(#) element $\xrightarrow{\text{Hash function}}$ key generate

Hashing.

key = Σ ascii value % 12 } → Hash function

Hash function, H = # of alphabets in a word/

(# of small, # of cap)

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ⓐ in / not in

membership operator

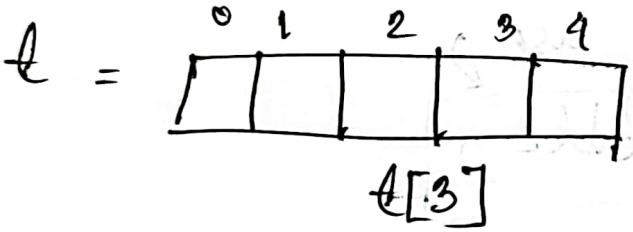
⇓ python

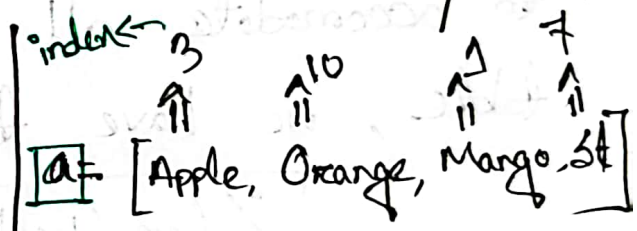
loop.

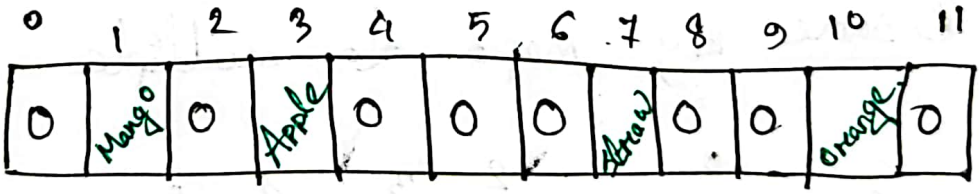
Hash table.

ⓐ We use another [array] for the mapping.

Array indexing access takes constant time / doesn't need a loop.







12
↳ size

Strawberry ⇒ 7

print (hash_table[7] == "Strawberry")

ⓐ asize = 10^7

to generate hash_table → 10^7 loop.

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size

(#) Hash table ⇒ 7

element in main array ⇒ ~~12~~ 9

a = [AB, CD, EF, CF, GH, IK, MN, ZF, FB]

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓

1 0 5 2 3 6 4 8 10.

Hash Function:

to accomodate all the elements within hash table, we have to use

`% len(hash table)`

(#) same key maps to multiple element.

↳ Collison

0	1	2	3	4	5	6
CD	AB	CF	GH	MN	EF	IK

h_t = ZF, FB.

Reasons behind collison

1) # of elements > size of Hashtable

2) # of elements < size of Hashtable.

Hash function isn't suitable.

Hash_function = $\sum$ ascii value / % len (hash_table)
serial of alphabet

a = [CaD, cAD, cad, CAD]
3 4

Collision Handling:

1) Linear Probing → Less used! → extremely problematic

2) Forward Chaining.

a = [CAD, PQR, ZAf, NZB]

ht =

	CAD	NZB	PQR	ZAF
0	1	2	3	4

↳ Linear Probing

Problem

1) size of array = size of ht

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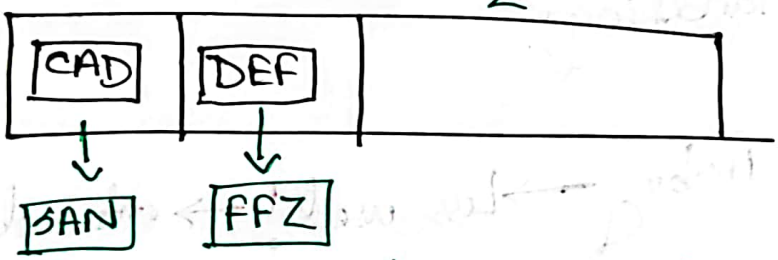
2) if not found, we have to move forward (loop)

Forward Chaining

a = [CAD, DEF, FFZ, SAN]

0 1 1 0

ht =



*linking → insertion at the very beginning.

1) size of array ≠ size of ht

2) linking in a single index

a = [ab, cd, ef, gh, ni, jk, zb, ji, ez]

0 1 1 0 2 0 0 0 0

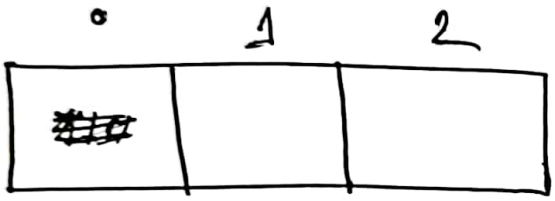
ht :

given_array = [z, c, nf, pq, to, ni, tw, fo]

keys : 0 1 2 0 1 1 2.

size of ht = 5

Draw the ht



Q-2

k	v
12	Apple
5	Orange
17	Banana
10	Grapes
22	watermelon
15	pineapple

Hash function : K%5

Generate Hash table.

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